Phase 1 Report: Data Audit and Descriptive Analysis of MKI67 Expression in TCGA-BRCA

Setayesh Hoseini
Statistics Course — Fall 1405

November 3, 2026

Abstract

This report summarizes Phase 1 of the project, which focuses on the data audit, cleaning, and descriptive analysis of the TCGA-BRCA dataset. The primary objective is to establish a robust, patient-level analytic dataset that links clinical tumor stage to RNA-seq gene expression. We processed 5,546 raw clinical records to isolate 1,083 unique patients with valid primary tumor RNA-seq data. Initial descriptive results suggest that MKI67 expression does not increase monotonically with AJCC stage, highlighting the need for rigorous inferential testing in Phase 3.

1 Introduction and Objectives

The goal of this project is to investigate whether MKI67 (a marker of cellular proliferation) mRNA expression is associated with AJCC pathologic tumor stage in breast cancer, and whether this association persists after adjusting for age at diagnosis. The unit of observation is strictly at the patient level. The analytic sample consists of 1,083 adult patients with primary invasive breast carcinoma who contributed both valid clinical data and a primary-tumor RNA-seq profile.

2 Data Preparation and Cleaning

The initial clinical dataset (`clinical.tsv`) contained 5,546 rows and 202 columns for 1,098 unique patients. The row explosion was structural, driven by multiple treatment records per patient.

2.1 Patient-Level Collapse

To ensure independence of observations, we applied the following rule:

1. Filtered for primary diagnoses (`diagnosis_is_primary_disease == "true"`), reducing to 4,946 rows.
2. Dropped all 44 `treatments.*` columns to prevent row duplication.
3. Collapsed duplicates based on `cases.case_id`, keeping the earliest diagnosis record.

This resulted in a clean table of 1,096 unique patients. Two patients were excluded due to missing primary-disease flags.

2.2 RNA-seq Mapping

From 1,231 RNA-seq files, we mapped sample barcodes to patient barcodes. We isolated primary tumor samples and excluded 11 patients with multiple primary tumors to maintain the "one-sample-per-patient" rule (Rule A). The final analytic sample after merging clinical and RNA-seq data is **1,083 patients**.

3 Descriptive Results

We extracted a pre-justified 6-gene panel (ESR1, PGR, ERBB2, MKI67, PCNA, GAPDH). GAPDH was used as a sanity check, with all values > 0, confirming RNA quality. Figure 1 shows the age distribution of the cohort.

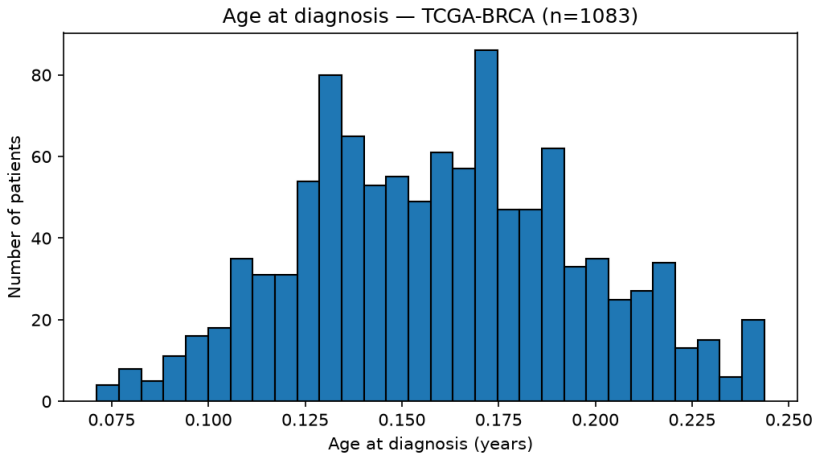


Figure 1: Age at diagnosis in the analytic sample.

Figure 2 shows the distribution of AJCC stages, heavily concentrated in Stage II.

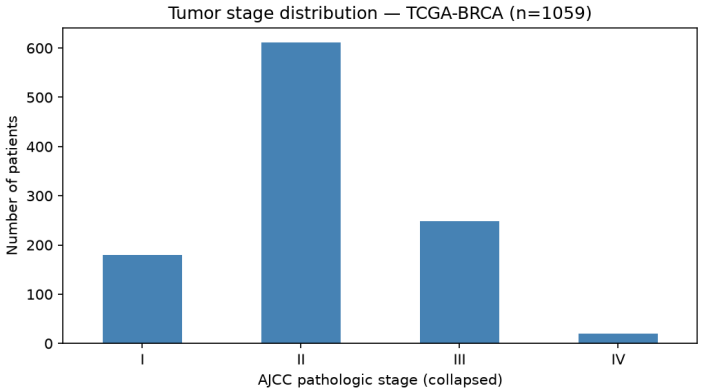


Figure 2: Distribution of AJCC pathologic stages (N=1,083).

Figure 3 displays the distribution of the 6-gene panel on the log scale, and Figure 4 demonstrates the right-skewed nature of raw MKI67 expression, justifying the $log(1+x)$ transformation for normality.

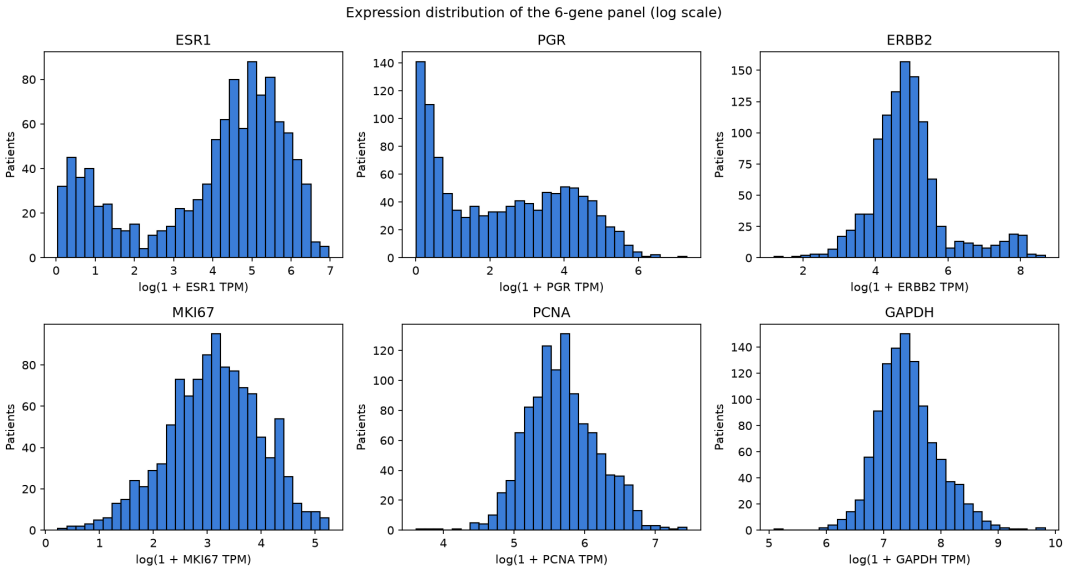


Figure 3: Expression distributions of the 6-gene panel (log scale).

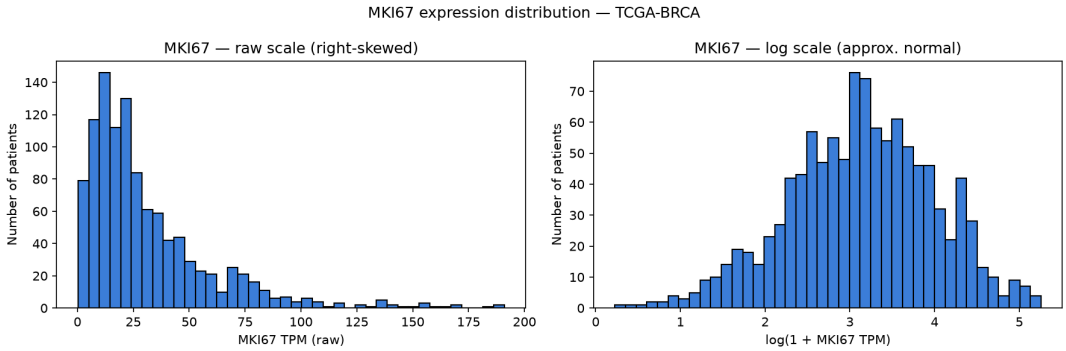


Figure 4: Raw vs. Log-transformed MKI67 expression. The log transformation corrects the skewness.

3.1 Group Comparison

Table 1 summarizes the log-transformed MKI67 expression by stage.

Table 1: Descriptive summary of log(MKI67) by stage (Early vs. Late).

Stage Group	N	Mean	Median	SD
Early (I–II)	791	3.145	3.159	0.852
Late (III–IV)	268	3.078	3.135	0.886

Figure 5 visually compares the distribution across groups. Notably, MKI67 expression does not increase monotonically with stage; early-stage tumors have a slightly higher mean expression than late-stage tumors.

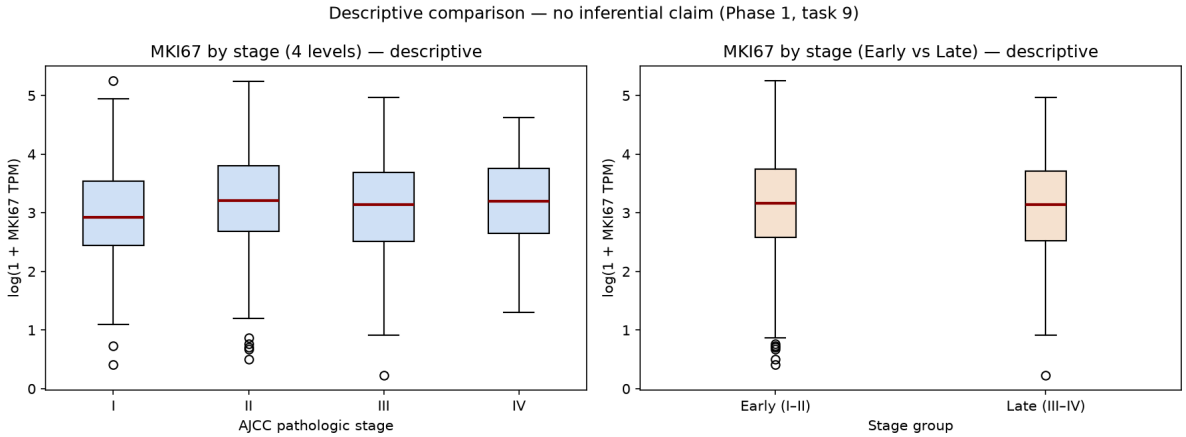


Figure 5: Boxplots of log(MKI67) by stage. No inferential claim is made in Phase 1.

4 Discussion and Limitations

While proliferation is a known driver of tumor aggressiveness, AJCC stage is predominantly determined by tumor size and lymph node involvement. The descriptive observation that MKI67 peaks in Stage II suggests that a simple monotonic relationship may not exist. Formal hypothesis testing (ANOVA and linear regression) will be conducted in Phase 3 and 4.

Limitations of this phase include the observational nature of the data (no causal claims), a small Stage IV subgroup ($N = 20$), and the fact that TCGA is a convenience sample, not a population-representative cohort.